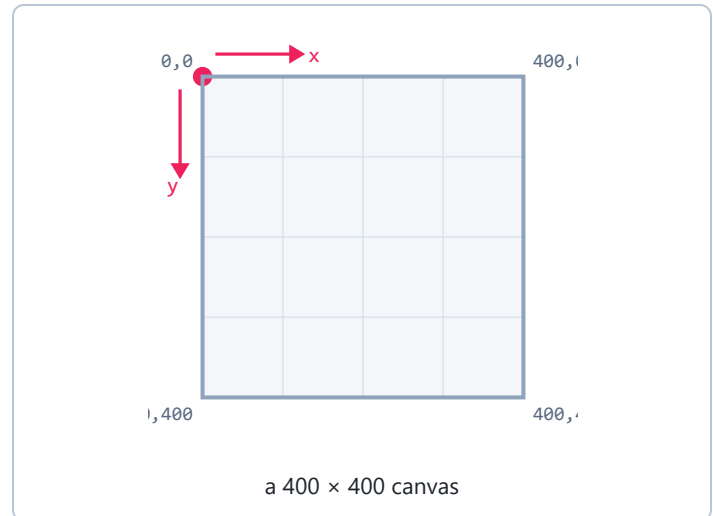


Where things are

Every point on the canvas is **two numbers**: x first, then y.

- x gets bigger going **right**
- y gets bigger going **down**

The origin (0, 0) is the **top-left corner**, not the middle. A screen fills in pixels starting from that corner, so that is where counting starts.



Every sketch has these two parts

```
function setup() {
  createCanvas(400, 400); // runs ONCE, at the start
}

function draw() {
  background(150); // runs over and over
  line(0, 0, 400, 400);
}
```

`createCanvas(400, 400)` is what makes `width` be 400 and `height` be 400. You never set them yourself — you just use them.

Drawing a line

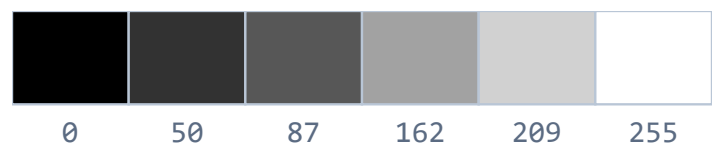
```
line(x1, y1, x2, y2);
//  \___/  \___/
//  start   end
```

Four numbers, but read them as **two points** — where the line starts and where it ends.

The four corners

top-left	0, 0
top-right	width, 0
bottom-left	0, height
bottom-right	width, height
the middle	width/2, height/2

Gray, from one number



`background(150)` the canvas
`stroke(0)` the line color
`strokeWeight(5)` how thick

Use width and height instead of typing numbers.

`line(0, 200, 400, 200)` and `line(0, height/2, width, height/2)` look identical on a 400×400 canvas. Change the canvas to 600×600 and only the second one is still in the middle.

Halves, thirds, quarters. `width/2` is the middle, `width/3` is one third across, `width*3/4` is three quarters. Coordinates can be worked out — they do not have to be typed.